11. Implement a simple top-down parser for context-free grammars using python.

Aim

To implement a simple top-down parser in Python to check whether an input string is accepted by a given context-free grammar (CFG).

Algorithm

1. Start the program.
2. Define the context-free grammar using production rules.
3. Read the input string.
4. Start parsing from the start symbol S.
5. Match the input symbols with the grammar rules.
6. If all input symbols are matched, accept the string; otherwise, reject it.
7. Display the result and stop.

Code :

```
# Simple Top-Down Parser

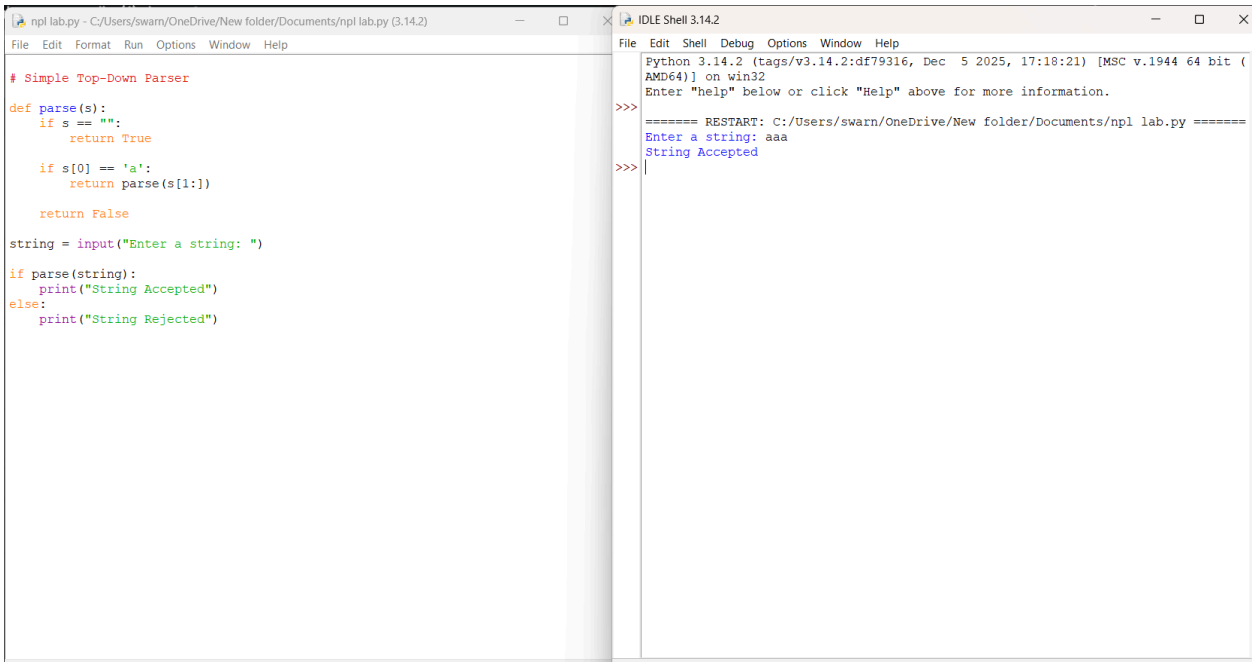
def parse(s):
    if s == "":
        return True

    if s[0] == 'a':
        return parse(s[1:])

    return False
```

```
string = input("Enter a string: ")


if parse(string):
    print("String Accepted")
else:
    print("String Rejected")
```



Result

Thus, a simple top-down parser was implemented successfully in Python to accept or reject strings according to the given context-free grammar.